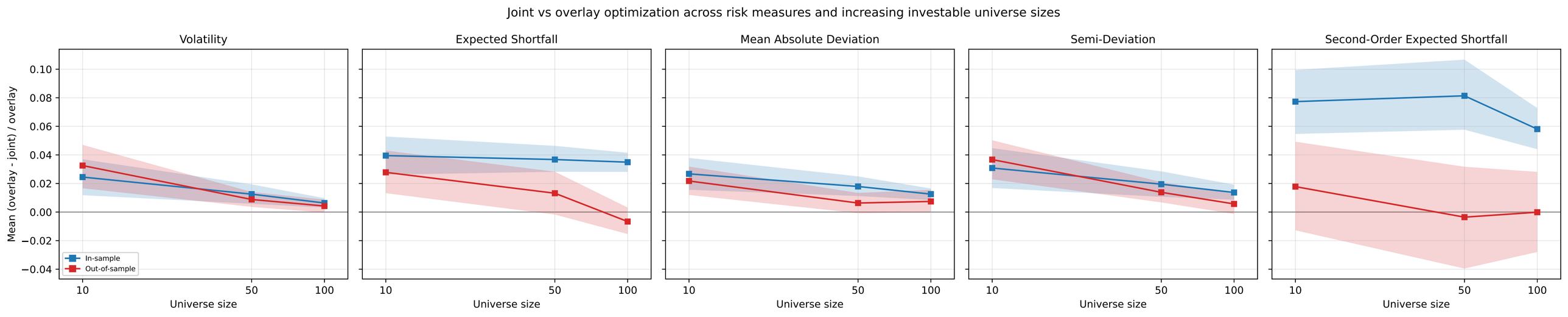
Joint vs overlay optimization across risk measures and increasing investable universe sizes
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Second-Order Expected Shortfall
Mean (overlay - joint) / overlay
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
In-sample
Out-of-sample
10
50
100
Universe size